Laxmi Karthikeya Gollapudi

glkarthik27@gmail.com | Portfolio | GitHub | Technical Writing

Summary

First Year Computer Science student at PES University with interests in machine learning, computer vision, and interdisciplinary applications of technology. Experience in developing projects involving real time sensing, data analysis, and machine learning models. Strong interest in research, technical writing, and exploring diverse areas of computer science through practical projects.

Education

PES University — Bengaluru, India
B.Tech in Computer Science and Engineering — *2025 – 2029*

Experience

ACM Applied Industrial Experience Program (AIEP) — Research Program
Machine Learning Research Participant — *Current*

- Participating in a collaborative research project comparing regularization techniques in machine learning models.
- Evaluating representation quality and generalization using PyTorch implementations.

Projects

ATLAS Tile Calorimeter Data Reconstruction | *Python, NumPy, Scikit-learn* — GitHub

- Implemented a linear regression model to reconstruct calorimeter energy signals.
- Evaluated reconstruction accuracy through signal analysis and benchmarking.

Anomaly Detection using prmon | *Python, PyTorch* — GitHub

- Generated synthetic monitoring datasets using prmon.
- Developed an autoencoder based anomaly detection model for abnormal resource usage.

Autoencoder Regularization Study | *Python, PyTorch, NumPy* — GitHub

- Implemented and compared multiple autoencoder architectures.
- Benchmarked models using MNIST for reconstruction accuracy and latent quality.

RPSGO Gesture Recognition Game | *YOLOv8, OpenCV* — GitHub

- Trained a YOLOv8 model for real time Rock Paper Scissors gesture recognition.
- Built automated scoring using OpenCV based hand detection.

Make-RB Pothole Detection App | *Kivy, Firebase* — GitHub

- Developed a mobile app using accelerometer and gyroscope data to detect road anomalies.
- Implemented filtering to distinguish potholes from normal motion.

Technical Skills

Languages: Python, C, HTML, CSS, JavaScript

Artificial Intelligence: Machine Learning, Deep Learning, Anomaly Detection, Computer Vision

Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn, Firebase, Kivy

Tools: Git/GitHub, Data Analysis, Technical Writing